



Immunology of IBD & Celiac Disease

Microbiology GIT Block · MEDucation by Mattin Majid

BEFORE YOU READ — ANSWER FROM MEMORY

1. Which antibody is more typical of ulcerative colitis, and which of Crohn's disease?
2. How does gluten trigger injury in celiac disease (enzyme, HLA, T-cell)?
3. Why must total IgA be checked when celiac serology is negative but suspicion is high?

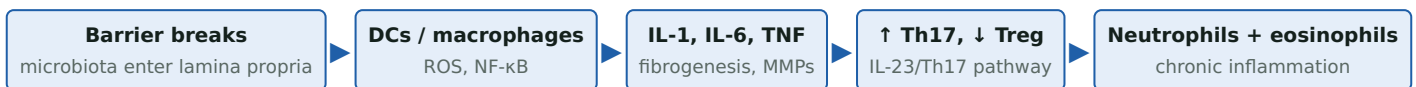
Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 IBD — the big picture

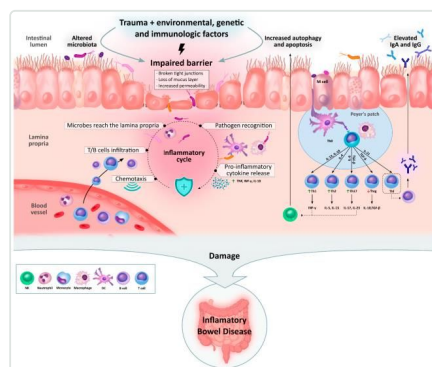
- **Immune-mediated, chronic, relapsing-remitting** inflammation; two types: **ulcerative colitis (UC)** and **Crohn's disease (CD)**; peak in the 2nd-3rd decade with a second peak in the elderly
- Causes: **genetics + barrier dysfunction + dysbiosis + dysregulated immunity + environment** (diet, air pollution, smoking, drugs, geography, stress); polygenic; higher risk in first-degree relatives and monozygotic twins
- Extra-intestinal: **seronegative arthritis, uveitis, sclerosing cholangitis**; malnutrition is common; complications: obstruction, perforation, fistulas, ↑ colorectal cancer risk

Gene / factor	Link
NOD2 / CARD15	Receptor for muramyl dipeptide of peptidoglycan — strongly associated with CD
HLA-DR*103	Severe UC and more extra-intestinal complications
IL-10, IL-10R, IL-23R	Implicated genes; CARD9 (neutrophils) protective against colitis
Diet / SCFA	High fat and sugar, low fibre → ↓ SCFA-producing bacteria (propionate, acetate, butyrate) and ↑ pathogens ; Mediterranean diet beneficial

2 Immunopathogenesis — the inflammatory cascade



Also: ILC3s major mediators of chronic inflammation; enhanced IgA against microbiota; B-cell perturbations.

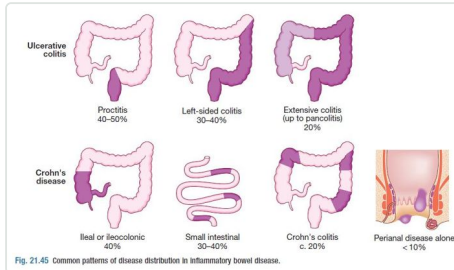


Overview of initiating inflammatory events in IBD.

3 Discriminator grid — ulcerative colitis vs Crohn's disease

	Ulcerative colitis	Crohn's disease
Distribution	Begins in the rectum , spreads continuously proximally, may involve whole colon	Any part of the GIT ; skip lesions
Lesion	Hemorrhagic, edematous, ulcerated mucosa; cryptitis and crypt abscesses ; pseudopolyps in long-standing disease	Focal inflammation, fistula tracts → fibrosis and stricture ; granulomas characteristic
Symptoms	Diarrhea, rectal bleeding, tenesmus , mucus, crampy pain	Abdominal pain, diarrhea, weight loss ; malabsorption when extensive
Complications	Toxic megacolon → perforation	Obstruction, fistulas; perforation 1–2% (usually ileum)

Smoking	More common in non- / ex-smokers (protective)	Risk factor
Antibodies	pANCA (pancolitis, primary sclerosing cholangitis); ASCA in 10–15%	ASCA in 60–70%; pANCA in CD → colonic disease
Labs	↑ CRP, platelets, ESR; ↓ Hb	↑ ESR, CRP; severe: hypoalbuminemia, anemia, leukocytosis
Other	HLA-DR*103 (severe)	<i>NOD2</i> ; low-residue, high-refined-sugar diet; cell-mediated response, defective T-suppressor function



Patterns of disease distribution in IBD.



Acute UC on colonoscopy.

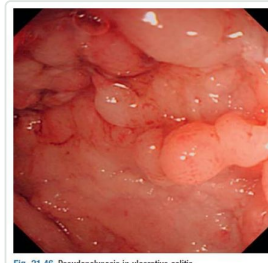


Fig. 21.46 Pseudopolyps in ulcerative colitis.

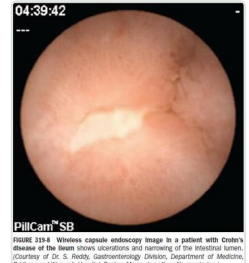


FIGURE 21.48 Wireless capsule endoscopy image in a patient with Crohn's disease of the ileum shows ulcerations and narrowing of the ileocecal junction. (Courtesy of Dr. S. Reilly, Gastroenterology Division, Department of Medicine, Brigham and Women's Hospital, Boston, Massachusetts, with permission.)

CD: ileal ulceration (capsule endoscopy).

Relapse triggers in UC: emotional stress, intercurrent infection, gastroenteritis, antibiotics, NSAIDs.

4 Markers and treatment logic

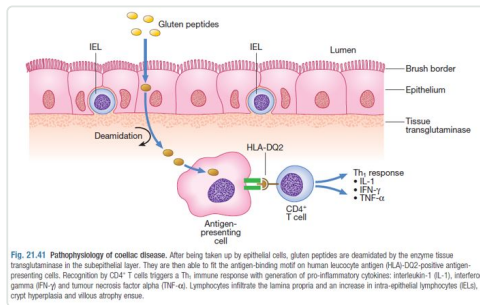
Item	Point
CRP	Elevated levels linked to more severe histological phenotypes
Fecal lactoferrin	High sensitivity/specificity to separate IBD from IBS
Calprotectin	Separates active from inactive IBD; monitoring tool
Biologics	Block TNF, IL-12, IL-23, IL-6, or leukocyte recruitment (anti-α4β7 integrin)
5-ASA and corticosteroids	Inhibit NF-κB and inflammatory mediators
Modern options	Probiotics, prebiotics, FMT

5 Celiac disease — illness script

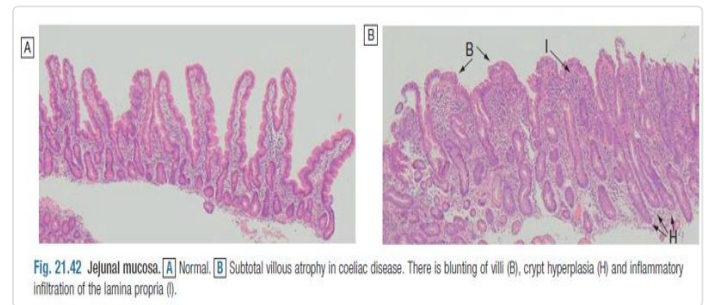
WHO	Genetically susceptible; HLA-DQ2 / DQ8 ; first-degree relatives and monozygotic twins; any age; " iceberg " disease. Associated: type 1 diabetes, IgA deficiency , Down syndrome, Addison's disease, thyroid disease
WHY	Immune reaction to gluten (gliadin) of wheat, rye, barley (less oats): tissue transglutaminase deamidates gliadin glutamine → new epitope binds DQ2 → IFN-γ Th1 CD4⁺ intraepithelial cells; IL-15 drives CD8 ⁺ , γδ T and NK cytotoxic killing of epithelium; B cells make anti-tTG and anti-gliadin / DGP antibodies
WHAT you see	Villous atrophy + crypt hyperplasia + lymphocytic infiltration. Infants : diarrhea, malabsorption, failure to thrive after weaning. Older children : delayed growth / puberty. Adults (3rd–4th decade, F:M 2:1): classic (diarrhea, pain, vomiting, fatty stools) vs atypical (anemia, osteopenia, infertility, neurologic) vs silent
WHERE it leads	2× malignancy risk (esp. enteropathy-associated T-cell lymphoma); some develop ulcerative jejuno-ileitis
NEVER miss	Total IgA : deficiency gives a false-negative IgA anti-tTG — test IgG antibodies in those patients

Diagnosis / treatment	Detail
Serology	IgA anti-tTG , anti-endomysial (EmA) , DGP ; titres fall on treatment; high-titre serology in children can be diagnostic without biopsy
Gold standard	Duodenal biopsy + positive serology ; biopsy when symptoms / labs suggest malabsorption plus positive tTG

Labs	Microcytic or macrocytic anemia (iron, folate); low vitamin D and other fat-soluble vitamins
Treatment	Lifelong gluten avoidance ; bowel returns to normal; antibodies vanish then reappear on gluten challenge; replace iron, folate, B ₁₂ , vitamin D



Pathophysiology of celiac disease.



Jejunal mucosa: normal vs subtotal villous atrophy.

P Must-remember pearls

- **UC** = rectum → continuous, cryptitis / crypt abscess, **pANCA**, protected by smoking. **CD** = skip lesions, granulomas, fistulae / strictures, **ASCA**, worsened by smoking.
- IBD immunology: barrier failure → ↑ **Th17**, ↓ **Treg**, IL-23/Th17, TNF/IL-1/IL-6 via NF-κB; **NOD2** = CD gene.
- **Fecal lactoferrin** separates IBD from IBS; **calprotectin** separates active from inactive.
- Celiac: **HLA-DQ2/DQ8**, **tTG deamidates gliadin** → Th1 IFN-γ; **villous atrophy + crypt hyperplasia**.
- Diagnose with **tTG-IgA / EmA + duodenal biopsy** (check **total IgA**); treat with **lifelong gluten-free diet**.

Q Test yourself

1. A 24-year-old non-smoker has bloody diarrhea and tenesmus; colonoscopy shows continuous rectal-to-colonic inflammation with crypt abscesses. Which antibody is most likely raised?

- A. ASCA
B. pANCA
C. Anti-tTG
D. Anti-mitochondrial

2. A smoker has abdominal pain, weight loss and fistulae; biopsy shows granulomas in an ileal segment with skip areas. Diagnosis?

- A. Ulcerative colitis
B. Crohn's disease
C. Celiac disease
D. Irritable bowel syndrome

3. A child has typical celiac symptoms but a low IgA anti-tTG. What should be checked?

- A. Total IgA; use IgG antibodies if deficient
B. Stool lactoferrin
C. pANCA only
D. Skin-prick tests

4. Which sequence explains celiac injury?

- A. tTG deamidates gliadin → binds DQ2 → Th1 IFN-γ response
B. Gliadin binds IgE → mast-cell degranulation
C. Lactase deficiency → osmotic diarrhea
D. NOD2 mutation → impaired peptidoglycan sensing

5. Which marker best separates IBD from irritable bowel syndrome?

- A. Fecal lactoferrin
B. Serum ferritin
C. Skin-prick test
D. Stool ova

ANSWERS — COVER UNTIL YOU'VE COMMITTED

- 1 — B** — pANCA is more common in UC; ASCA in CD.
2 — B — Skip lesions, granulomas and fistulae with smoking indicate CD.
3 — A — IgA deficiency is frequent in celiac disease and gives false-negative IgA tests.
4 — A — Deamidated gliadin binds HLA-DQ2 and drives IFN-γ Th1 cells.
5 — A — Elevated fecal lactoferrin is a sensitive and specific marker of IBD inflammation.

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam